

PL/SQL Complete Notes for Interview Preparation

1. Introduction to PL/SQL

PL/SQL is Oracle's procedural extension of SQL. It allows you to write logic like loops, conditions, and functions.

2. PL/SQL Block Structure

A PL/SQL block has DECLARE, BEGIN, EXCEPTION, and END sections.

3. Variables & Data Types

Common types: NUMBER, VARCHAR2, DATE. Use %TYPE and %ROWTYPE for dynamic typing.

4. Control Statements

Includes IF-ELSE, CASE, LOOP, WHILE LOOP, FOR LOOP.

5. Cursors

Used to handle multiple rows. Types: Implicit and Explicit Cursor.

6. Exception Handling

Handles runtime errors like NO_DATA_FOUND, TOO_MANY_ROWS.

7. Stored Procedures

Reusable block of code stored in database.

8. Functions

Returns a value and can be used in SQL queries.

9. Triggers

Automatically executed on events like INSERT, UPDATE, DELETE.

10. Packages

Group of procedures and functions for better structure.

11. Collections

Advanced structures: VARRAY, Nested Tables, Associative Arrays.

12. Bulk Operations

BULK COLLECT and FORALL improve performance.

13. Dynamic SQL

EXECUTE IMMEDIATE allows runtime SQL execution.

Interview Questions

1. What is PL/SQL? 2. Difference between SQL and PL/SQL? 3. What is a cursor? 4. Difference between procedure and function? 5. What is a trigger? 6. What is exception handling?